



NCI Protocol #: 10323

Patient ID:

Date:

1 of 6

Patient Name:

Date Collected:

Biopsy Site: **Bone Marrow Aspirate (Iliac Crest)**

Sample ID:

Telephone:

Tumor Content (%): **0%**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Acute Myeloid Leukemia**

MoCha ID:

Fax:

Preanalytics to include tumor enrichment as required is performed by Roswell Park Comprehensive Cancer Center; Buffalo; NY 14203

Relevant Acute Myeloid Leukemia Findings

Gene	Finding	Gene	Finding
ABL1	None detected	MECOM	None detected
ASXL1	None detected	MLLT3	None detected
CEBPA	None detected	MYH11	None detected
CREBBP	None detected	NPM1	NPM1 W288Cfs*12
FLT3	None detected	NUP214	None detected
IDH1	None detected	RARA	None detected
IDH2	None detected	RUNX1	None detected
KMT2A	None detected	TP53	None detected

Relevant Biomarkers

No biomarkers associated with relevant evidence found in this sample

Prevalent cancer biomarkers without relevant evidence based on included data sources

PTPN11 G503A, TET2 W1847, TET2 R544*, NPM1 W288Cfs*12*

Variant Details

DNA Sequence Variants

Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Coverage	Transcript	Variant Effect
PTPN11	p.(G503A)	c.1508G>C	COSM13027	chr12:112926888	7.25%	2000	NM_002834.5	missense
TET2	p.(R544*)	c.1630C>T	COSM41850	chr4:106156729	7.32%	1818	NM_001127208.2	nonsense
TET2	p.(W1847*)	c.5541G>A	.	chr4:106197208	9.38%	1845	NM_001127208.2	nonsense
NPM1	p.(W288Cfs*12)	c.863_864insTCTG	COSM17559	chr5:170837547	6.76%	1967	NM_002520.6	frameshift Insertion

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Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2023.03(005). The content of this report has not been evaluated or approved by FDA, EMA or other regulatory agencies.

Low Coverage Regions:

N/A

Comments:

The above listed mutations were also identified in previously tested pre-treatment diagnostic specimen of this patient, albeit, at much higher variant allele frequencies (OCA-01291, reported on 02/21/2023). These findings may indicate reduced disease burden in currently obtained post induction bone marrow.

Report Signed By

Name	Role	Date

Biomarker Descriptions

NPM1 (nucleophosmin 1)

Background: The NPM1 gene encodes the nucleophosmin protein, a histone chaperone of the nucleophosmin/nucleoplasmin family, which also includes NPM2 and NPM3¹. NPM1 functions as an oncogene and tumor suppressor, and is important in maintaining genomic stability, DNA repair, and apoptosis^{1,2}. NPM1 has a highly conserved N-terminal region which constitutes the core domain responsible for oligomerization, an acidic domain, a nuclear localization signal, and a disorganized C-terminal region which is required for nucleolar localization¹. Oligomerization of NPM1 localizes the protein in the nucleus of proliferating cells where it binds to Akt in response to growth factor stimulation and escapes proteolytic degradation by caspase activity, thereby promoting cell survival^{1,2}. NPM1 is one of the most frequently altered genes in hematological cancers³. Most NPM1 mutations occur in the C-terminus, impacting protein folding or the nucleolar localization signal, and result in the localization of NPM1 to the cytoplasm (NPMc) instead of to the nucleus¹.

Alterations and prevalence: NPM1 mutations are observed in 45-60% of AML with a normal karyotype (NK-AML), 28-35% of de novo acute myeloid leukemia (AML) and are frequently co-mutated with DNMT3A and/or FLT3-ITD^{4,5,6}. NPM1 fusions are associated with distinct partner genes in acute promyelocytic leukemia (APL), anaplastic large-cell lymphoma (ALCL), AML, and myelodysplasia³. Specifically, NPM1-ALK fusion is found in 30% of all ALCL and this specific fusion is observed in 85% of ALK-positive ALCL¹. The t(5;17) (q35;q21) translocation that results in NPM1-RARA fusion is observed in APL⁷.

Potential relevance: Mutation of NPM1 is recognized as a diagnostic entity for AML with NPM1 mutation by the World Health Organization (WHO)⁸. NPM1 mutations are associated with better outcomes, increased complete remission, improved overall survival, and favorable risk in AML^{4,6,9}. Concurrent expression of FLT-ITD with mutant or wild-type NPM1 (when lacking adverse risk genetic lesions) confers intermediate risk in AML^{4,9}. The NPM1 frameshift mutation W288fs*12 is associated with poor prognosis in myelodysplastic syndrome (MDS)¹⁰. The ALK-NPM1 fusion and translocation t(2;5)(p23;q35) which leads to an ALK-NPM1 fusion is diagnostic of ALK-positive anaplastic large cell lymphoma^{11,12}.

PTPN11 (protein tyrosine phosphatase non-receptor type 11)

Background: The PTPN11 gene encodes a tyrosine phosphatase non-receptor type 11 protein, and is also known as Src homology region 2 domain-containing phosphatase-2 (SHP-2)¹³. PTPN11 is a member of the protein tyrosine phosphatase (PTP) family that is ubiquitously expressed and regulates cellular growth, differentiation, mitotic cycle, and oncogenic transformation. PTPN11 contains two tandem N-terminal Src homology-2 domains (N-SH2 and C-SH2), a PTP catalytic domain, and uncharacterized C-terminal domain¹⁴. PTPN11 regulates various signaling processes including the RAS/RAF/MEK/ERK, PI3K/AKT/MTOR, and JAK/STAT pathways^{15,16}. Germline mutations in PTPN11 are associated with LEOPARD syndrome and Noonan syndrome with a predisposition to juvenile

Biomarker Descriptions (continued)

myelomonocytic leukemia (JMML) or myeloproliferative neoplasms (MPN)^{10,17}. Somatic mutations in PTPN11 are associated with JMML^{18,19} and solid tumors such as lung, colon, and thyroid^{14,20}

Alterations and prevalence: Somatic alterations in PTPN11 include mutations and amplification^{17,21}. PTPN11 mutations occur in 6% of uterine carcinoma and 5% of acute myeloid leukemia (AML) cases²². Mutations including E76K and D61Y result in PTPN11 activation and are associated with 30% of JMML¹⁶.

Potential relevance: Currently, no therapies are approved for PTPN11 aberrations. Somatic mutations in PTPN11 confer drug resistance to venetoclax and azacitidine in AML^{23,24}.

TET2 (tet methylcytosine dioxygenase 2)

Background: TET2 encodes the tet methylcytosine dioxygenase 2 protein and belongs to a family of ten-eleven translocation (TET) proteins that also includes TET1 and TET3²⁵. TET2 is involved in DNA methylation, specifically in the conversion of 5-methylcytosine to 5-hydroxymethylcytosine^{26,27}. The TET proteins contain a C-terminal core catalytic domain that contains a cysteine-rich domain and a double stranded β -helix domain (DSBH)²⁸. TET2 is a tumor suppressor gene. Loss of function mutations in TET2 are associated with loss of catalytic activity and transformation to hematological malignancies^{25,26,27}

Alterations and prevalence: Somatic TET2 mutations, including nonsense, frameshift, splice site, and missense, are observed in 20-25% of myelodysplastic syndrome (MDS) associated diseases, including 40%-60% chronic myelomonocytic leukemia (CMML)¹⁰. TET2 mutations at H1881 and R1896 are frequently observed in myeloid malignancies^{26,29}. TET2 mutations are also observed in 9% of uterine, 8% of melanoma and acute myeloid leukemia (AML), as well as 6% of diffuse large B-cell lymphoma (DLBCL).

Potential relevance: The presence of TET2 mutations may be used as one of the major diagnostic criteria in pre-primary myelofibrosis (pre-PMF) and overt PMF in the absence of JAK2/CALR/MPL mutations^{30,31}. TET2 mutations are associated with poor prognosis in PMF and increased rate of transformation to leukemia^{31,32}

Assay Information

Methodology, Scope, and Application: The NCI Myeloid Assay v2 (NMAv2) is a next-generation sequencing (NGS) assay which identifies pre-defined and novel genomic variants covering 45 DNA genes and 35 fusion drivers that are categorized into 3 mutation types: single nucleotide variants (SNVs), insertions/deletions (Indels) including FLT3 internal tandem duplications (ITDs), and gene fusions. The assay utilizes Thermo Fisher Scientific's Oncomine® Myeloid Assay GX v2, a NGS assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from blood and bone marrow mononuclear cells for sequencing on the Ion Torrent Genexus Integrated Sequencer and analyzed by the Ion Torrent Genexus pipeline version 6.6.2.1. The NMAv2 can reliably identify the presence or absence of 1661 hotspot variants and 779 reported fusions compared to the Human Reference Genome hg19, including the genes listed below. The NMAv2 is a laboratory developed test designed to detect gene mutations for major myeloid disorders.

Analytical Sensitivity and Specificity: The NMAv2 has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 98.97% sensitivity, 100% specificity and an overall reproducibility with a mean positive percent agreement of 99% and a mean negative percent agreement of 100% during validation of the assay.

Hereditary and Germline Mutations: This assay examines cancer cells only and does not examine normal (non-cancer) cells. Mutations detected by the assay may be present only in the cancer cells, or in every cell of the body (including non-cancer cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or more features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's), it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-cancer) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent OncoPrint Knowledgebase Reporter (OKR) software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OKR. The data presented here is from a curated knowledgebase of publicly available information but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

ABL1, ANKRD26, BRAF, CBL, CSF3R, DDX41, DNMT3A, FLT3, GATA2, HRAS, IDH1, IDH2, JAK2, KIT, KRAS, MPL, MYD88, NPM1, NRAS, PPM1D, PTPN11, SETBP1, SF3B1, SMC1A, SMC3, SRSF2, U2AF1, WT1

Genes assayed with full exon coverage:

ASXL1, BCOR, CALR, CEBPA, ETV6, EZH2, IKZF1, NF1, PHF6, PRPF8, RB1, RUNX1, SH2B3, STAG2, TET2, TP53, ZRSR2

Genes assayed for detection of fusions:

ABL1, ABL2, BCL2, BRAF, CCND1, CREBBP, EGFR, ETV6, FGFR1, FGFR2, FUS, HMGA2, JAK2, KAT6A, KAT6B, KMT2A, KMT2A-PTD (Partial Tandem Duplication), MECOM, MET, MLLT10, MRTFA (MKL1), MYBL1, MYH11, NTRK2, NTRK3, NUP214, NUP98, PAX5, PDGFRA, PDGFRB, RARA, RUNX1, TCF3, TFE3, ZNF384

Limitations of the assay

1. Mutations in homopolymer regions of >5 nucleotides may not be accurately assessed by this assay.
2. FLT3-ITDs greater than 120bps may not be detected by this assay. Sensitivity for larger FLT3-ITD's may be reduced and variant allele frequencies may be reported lower than expected.
3. Following genetic alterations will not be reported by this assay: RUNX1 p.Ser445Ala/Trp149, ASXL1 p.G646Wfs*12, PRPF8 p.Arg156Glu/Trp28

DISCLAIMER

This assay reports SNV, indel and FLT3-ITD mutations $\geq 5\%$ Variant Allele Frequency (VAF). This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

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